I. INTRODUCTION:

Search engines are among the most influential technologies of the modern era, acting as primary gateways to information, education, and decision making. What began as simple keyword-matching systems in the 1990s has evolved into sophisticated, AI-driven platforms capable of understanding natural language, generating contextual responses, and personalizing results based on user behavior. This evolution represents not just incremental technical improvements, but a completely new concept of how users interact with information and how they perceive the authority and trustworthiness of information sources.

Traditional search engines established certain user experience patterns and expectations. Users would enter keywords, receive a ranked list of relevant web pages, evaluate these sources independently, and navigate to the most promising documents. This interaction model empowered users to act as critical evaluators of information quality, credibility, and relevance. Users developed mental models of how search worked, which shaped their trust in search results and their information-seeking strategies. The relationship between user and search engine was relatively clear. The engine served as a mediator between the user and available documents, not as an authority itself.

The integration of artificial intelligence has totally altered this user experience and the nature of the user-search engine relationship. Modern AI-enhanced search engines like Google Search with AI Overviews, Bing Copilot, and Perplexity AI no longer simply return lists of links for users to explore. Instead, they interpret user intent, generate synthesized answers, engage in conversational exchanges, and provide tailored summaries. Users now frequently receive direct answers rather than source lists, engaging with AI-generated content that may synthesize information from documents they never directly access. This shift promises greater efficiency and accessibility, but also introduces significant questions about user trust, information literacy, and critical evaluation skills.

This Honors Thesis investigates how the integration of artificial intelligence has reshaped search engine functionality, effectiveness, and user experience when compared to traditional, non-AI systems. Working collaboratively with Shahab Kiyani, who focuses on the technical implementation and system-level analysis, my contribution centers on understanding how users perceive, trust, and interact with these different search models. While Shahab designed and implemented a custom traditional search engine to enable rigorous technical comparison, my work examines how real users experience both traditional and AI-enhanced search systems in practice.

My research employs a survey-based methodology to capture user perceptions, preferences, and concerns. Due to practical constraints, responses were generated through controlled evaluations conducted by both researchers, alongside additional responses produced by an AI system, allowing for consistent comparison across conditions. I gathered data on perceived accuracy, trustworthiness, usability, satisfaction with results, an understanding of how systems work, confidence in verifying information, and preferences for different types of information needs. This user-centered research complements Shahab's technical analysis, providing essential insights into whether technical improvements in AI systems translate into meaningful enhancements in user experience. Together, these dual perspectives provide clearer insights into whether AI-powered search systems meaningfully improve information access or just introduce new risks to information literacy and informed decision-making.

A. Scientific and Intellectual Problem and Objectives:

The central focus of this thesis is to examine how AI-integrated search engines differ from traditional search systems in terms of user experience, trust, and information-seeking behavior. Rather than attempting to definitively determine whether one approach is superior, this work is motivated by the bigger scope intellectual problem of understanding the tradeoffs introduced by AI-driven search. These tradeoffs include potential improvements in efficiency and accessibility alongside concerns such as reduced transparency, decreased critical evaluation, and misplaced user trust. As search engines continue to play a central role in decisions ranging from academic research to everyday problem solving, exploring how these systems shape user interaction remains an important and timely area of study.

Traditional search engines established user experience standards that many stakeholders valued. Users received reliable lists of available sources, allowing them to see a wide range of viewpoints, not just one answer. Results were presented with clear signs of where they came from including URLs, page titles, and brief excerpts. This enabled users to make informed decisions about which sources to trust before clicking. Traditional search maintained a clear boundary between the search platform and the information it retrieved, encouraging users to actively evaluate credibility and engage in critical thinking when interpreting results.

AI-enhanced search systems challenge these established interaction patterns. When systems generate direct answers, they shift from acting as intermediaries to appearing as authoritative sources. Users may rely on AI-generated summaries without engaging with any underlying sources, potentially reducing opportunities for credibility evaluation. The complexity of these systems can make it difficult to understand how answers are produced, limiting transparency and user control. Additionally, AI systems may generate plausible but incorrect information, introducing new risks that are not always immediately detectable.

To examine these differences, this research uses a structured evaluation framework that focuses on direct comparison across search systems. Instead of emphasizing data collection alone, this approach centers on how responses vary depending on the system used and how those differences influence interpretation and decision-making. Evaluations are carried out through repeated trials conducted by the researchers, along with additional outputs generated by an AI system, creating a consistent setting for comparison. While this method has limitations, including potential bias constraints on how widely these results can be applied, it allows for a closer look at how each system shapes the presentation of information and the way it is understood. Ultimately, this process is intended to highlight practical differences in how traditional retrieval-based systems and AI-generated responses influence user experience, while reinforcing key concepts explored throughout this study.

B. Primary Objectives and Research Questions:

The main objective of this research is to demonstrate a proof-of-learning approach by applying what we've learned about both traditional search systems and AI-enhanced search in a practical setting. Instead of focusing on large-scale studies or trying to prove one system is definitively better, this work is centered on building, testing, and comparing these systems to better understand how they actually behave. The goal is to show a clear understanding of key concepts from information retrieval and AI by directly working with them and evaluating how they perform in a controlled setup.

Beyond the comparison itself, this project is also meant to reflect the learning process that comes from working hands-on with these systems. By designing, testing, and analyzing both traditional and AI-based approaches, we are able to better understand their strengths, limitations, and real-world tradeoffs. This experience goes beyond theory, helping us build a more direct understanding of how these technologies function and how they impact user interaction. The objective is not just to compare systems, but to show growth in understanding through direct application and exploration within this field.

Guided by these objectives, this study explores several key questions related to how search systems shape user experience. It examines how traditional and AI-enhanced search engines differ in the way information is presented and interpreted, particularly across dimensions such as trust, accuracy, and usability. It also considers the differences between human-generated and AI-generated responses when interacting with these systems, and how these differences affect perceived understanding and confidence in results. Additionally, this work investigates how the shift from document retrieval to AI-generated answers influences transparency, verifiability, and the tendency to engage in deeper source evaluation. Through this analysis, the study seeks to identify the tradeoffs between efficiency, accessibility, and critical engagement that emerge when comparing traditional and AI-enhanced search approaches within a controlled evaluation setting.

C. Significance of the Study:

From a human-centered perspective, this research looks at how changes in search technology can affect the way people find and understand information. As AI-powered search tools become more common in everyday life, from schoolwork to general problem solving, it becomes important to think about how these systems influence trust, understanding, and decision-making. This project does not aim to make broad claims about all users, but instead focuses on exploring these ideas through direct interaction with both traditional and AI-based search systems. By working hands-on with these technologies, this study helps highlight how differences in system design can impact the way information is presented and interpreted.

The importance of this work also comes from the learning process behind it. Through building, testing, and comparing these systems, we gain a better understanding of how search engines actually function beyond just theory. This includes recognizing both the strengths and limitations of AI-generated answers compared to traditional results. While the findings are based on a controlled evaluation, they still provide useful insight into how features like direct answers, source visibility, and system transparency can influence user experience. This kind of hands-on exploration is valuable not only for understanding current technologies, but also for thinking more critically about how they should be used or built in the future.

In addition, this project helps connect technical concepts with real-world implications. Even without large-scale user data, examining how these systems behave in practice can reveal important tradeoffs, such as convenience versus transparency or speed versus verifiability. These insights are not just about which system performs better, but about how people interact with information itself. As search tools continue to evolve, the question becomes less about what information is available and more about how that information is delivered and trusted. In that sense, the significance of this work is not only in comparing systems, but in recognizing that the way answers are presented can shape how people think, what they believe, and how they make decisions.

<u>II. SUMMARY OF WORK OF PREVIOUS RESEARCHERS</u>:

To better understand how search has changed and how people interact with these systems, it's important to look at what existing research has already found. Prior studies help explain not only how traditional and AI-based search systems work, but also how users experience them in terms of trust, understanding, and overall usability. By building on this research, this study focuses on how users actually experience the differences between traditional and AI-based search systems, while also addressing the lack of direct, side-by-side comparisons between them in a controlled setting.

The rapid transformation of search technologies has generated significant scholarly attention across disciplines including information retrieval, human computer interaction, marketing, and artificial intelligence research. Scholars have increasingly sought to understand not only how search systems function technically but also how users perceive, trust, and interact with them.

This body of literature provides the intellectual foundation for the present study by demonstrating that search engines are not merely technical tools but complex social systems in the sense of how they shape knowledge acquisition, decision making, and everyday behavior. Researchers have written extensively about the evolution from traditional keyword based retrieval systems to AI enhanced conversational platforms, highlighting the importance of user attitudes, experience, transparency, and psychological relationships with technology. Among the major influencers in this field are researchers who focus on information retrieval theory and user behavior, particularly William Hersh, Shu Sheng Liaw, and Hsiu Mei Huang. Their work highlights the enduring relevance of search engines while also acknowledging the disruptive potential of generative artificial intelligence. At the same time, newer research by Kim and Priluck expands the conversation into consumer psychology, while broader human centered AI frameworks presented by Shin and colleagues connect technological development to user well being. Together, these scholars provide both the historical grounding and the modern context that inspire the present research.

Early literature established the fundamental importance of search engines in navigating the vast amount of information available online. Liaw and Huang explain that "search engines are essential tools for finding information on the World Wide Web" (Liaw & Huang, 2011). This statement captures a core assumption that continues to guide modern research. Without structured retrieval tools, the overwhelming volume of digital information would make it nearly impossible to access the preferred documents. The authors further emphasize the difficulty users face when attempting to locate relevant material, noting that "retrieving relevant or needed information is far from assured" due to the enormous growth of online content (Liaw & Huang, 2011). These insights align closely with the motivation presented in this thesis, which frames search engines as primary gateways to knowledge. The early user focused approach introduced by Liaw and Huang is particularly influential because it shifts the analytical lens from system performance to human behavior. Rather than measuring only algorithmic efficiency, this perspective recognizes that technology succeeds only when users can effectively interact with it. Their findings demonstrate that "experience with search engines significantly affects user's attitudes toward search engines for information retrieval" (Liaw & Huang, 2011). This idea is especially relevant to the present project, which investigates perception and trust across different search models. If user experience shapes attitudes, then the introduction of AI driven interfaces likely alters how individuals evaluate search results. The authors also highlight limitations in traditional systems. They report that “users are not very satisfied with the precision of retrieved information of search engines and their response time” (Liaw & Huang, 2011). This dissatisfaction helps explain why innovation in search has continued. Users demand faster responses and more accurate results, pressures that have contributed to the integration of machine learning and AI technologies. Yet not being satisfied does not directly mean rejection. Instead, it signals an ongoing tension between technological capability and user expectations as well as the race to further improve and optimize newer systems regarding this tension.

Another important contribution from this study concerns information literacy. The authors argue that “electronic searching skills must be learned” because evaluating and filtering sources requires strong capability (Liaw & Huang, 2011). This observation directly informs the current thesis by suggesting that technological change may reshape not only search interfaces but also the cognitive skills required to navigate them. If AI systems simplify the search process, users may rely less on these skills, raising questions about critical evaluation and trust.

While foundational research focused heavily on traditional systems, current studies examine how generative AI has disturbed established models. Hersh provides one of the clearest understandings of this shift, writing that “information retrieval systems are widely used tools for information seeking in biomedicine and health and just about all other aspects of our lives” (Hersh, 2024). This statement reinforces how universal search technology is and stresses why studying its evolution matters. However, Hersh also notes a dramatic turning point, explaining that search systems “had been relatively mature applications until late 2022, when any staidness of search systems was upended by the emergence of generally available generative artificial intelligence chatbots” (Hersh, 2024). This transformation mirrors the central problem of the thesis, which asks whether AI represents real advancement or introduces new risks. Despite the excitement surrounding AI tools, Hersh expresses caution about their limitations. He observes that generative chatbots often provide outputs where “the user has no idea what actual source text led the generative model to output the specific text” (Hersh, 2024). This lack of traceability directly relates to concerns about transparency and verifiability discussed earlier in this thesis. When users cannot identify sources, trust becomes much more complicated. Hersh further emphasizes the importance of credibility in academic searching, stating that “one of the first things I look at when retrieving something is who wrote it and who published it” (Hersh, 2024). This perspective highlights a key difference between traditional search and AI generated responses. Traditional engines tell you exactly who wrote something and where it came from. AI chatbots usually hide that information. The persistence of traditional search behavior is another notable theme. Hersh explains that he still “heads first to search engines over generative AI chatbots for information seeking” because original sources remain more valuable for scholarly work (Hersh, 2024, pg 7). This insight suggests that AI has not fully replaced established methods, reinforcing the need for comparative research such as this study.

Building upon this perspective, consumer behavior research by Kim and Priluck introduces psychological dimensions that go beyond usability. Their work explains that traditional search engines “provide lists of links, AI chatbots have a natural conversational interface, which allows for detailed and direct interactive results” (Kim & Priluck, 2025). This difference in interaction style represents more than a technical change. It alters how users engage with information entirely. Surprisingly, the authors find that “consumers prefer search engines over AI chatbots and are more willing to self disclose their personal information to search engines due to perceived familiarity” (Kim & Priluck, 2025). Familiarity appears to generate comfort, suggesting that long standing technologies benefit from accumulated trust which makes sense for anything, not just search engines. Yet the relationship is not entirely straightforward. The study also reports that consumers “evaluate AI chatbot based results as less biased than traditional search engine results” (Kim & Priluck, 2025). This finding introduces an important contradiction. Users may trust search engines because they are familiar, but they may simultaneously suspect commercial bias within them. AI systems therefore represent a middle ground position, appearing novel and potentially more neutral. The authors further explain that conversational features “disrupt the sequential search mode of traditional search engines” while traditional platforms remain dominant (Kim & Priluck, 2025). This disturbance reflects the broader technological shift identified throughout the literature. AI does not merely enhance search. It challenges the very structure and process of information seeking.

Additionally, user experience design has emerged as a central focus in recent research on search technologies. The conference work by Šolar highlights the growing relevance of AI assisted interfaces, noting that classic engines rely on similarity based algorithms while newer systems combine these with conversational methods. The introduction of chatbots changes how users interpret machines, as studies suggest that “humans tend to perceive them as more humanlike, changing the UX” (Šolar, 2023). This perception is critical because humanlike interaction may increase engagement while simultaneously blurring the boundary between tool and authority. Šolar’s research seeks to examine “how the UX in regard to interaction and interface design differs between the classic and AI assisted browser search engines” (Šolar, 2023). This question closely aligns with the current thesis and demonstrates that scholarly interest is shifting toward experiential comparisons rather than purely technical evaluations. Collectively, these sources reveal several patterns that guide the present research. First off, search engines remain foundational to digital life despite rapid technological change. Secondly, user experience strongly shapes attitudes toward technology. Lastly, AI creates new ways for us to interact with technology, making it harder to know how decisions are made and who is in charge.

The scholars discussed above serve as inspirations for this project because they collectively frame search as a human centered activity rather than a purely computational process. Liaw and Huang inspire the emphasis on user attitudes. Hersh motivates critical examination of generative AI. Kim and Priluck encourage attention to psychological relationships with technology. Šolar highlights the importance of interface design. Their work demonstrates that understanding search requires integrating perspectives from information science, psychology, and human computer interaction. This literature also clarifies the gap that the current study aims to address. While researchers have explored user attitudes toward traditional search and emerging AI tools separately, fewer studies directly compare how users perceive both systems within the same evaluative framework. By building on these foundational works, this thesis seeks to contribute empirical evidence about trust, usability, and perceived accuracy across multiple search models. As artificial intelligence becomes more integrated into everyday technologies, scholars have increasingly focused on the expanded implications of human AI interaction. While early research emphasized retrieval accuracy and system performance, modern literature expands this conversation to include trust, transparency, cognitive understanding, and long term user well being. These themes are particularly relevant to this thesis because they help explain not only how users interact with search systems but also why certain technologies are trusted more than others.

One of the most influential perspectives in this space is the growing emphasis on human centered AI design. Shin and colleagues argue that current research is divided across fields and so it requires a more consolidated view. Their study states that it “aims to develop a comprehensive model for designing human centered AI systems that promote digital well being” (Shin, 2025). This effort reflects a broader movement within computing research that prioritizes the human experience rather than technological advancement alone. The authors further explain that their review produced “a conceptual framework that consolidates current knowledge on designing AI systems to support digital well being and positively influence human behavior” (Shin, 2025). This framework is especially meaningful for research on search engines because these platforms shape daily habits of information consumption. If AI driven systems influence emotions, perceptions, and behaviors, then evaluating their user experience becomes essential. Shin further highlights that digital well being concerns move past usability. Researchers have noted that excessive interaction with computing systems “could result in physical or mental health problems for users” (Shin, 2025). This observation broadens the stakes of search technology. What might appear to be a simple interface decision could ultimately affect cognitive load, attention, and overall health. The study then outlines a major research gap, explaining that there is “still a lack of studies that systematically reframe the concept of digital well being with AI powered computing systems” (Shin, 2025). This absence implies the need of ongoing investigation into how AI mediated information environments shape user behavior. This thesis contributes to this area by evaluating trust and perception within AI enhanced search contexts.

Closely related to well being is the concept of transparency, which has become a central concern in human AI interaction research. Liao, Varshney, and Weld describe transparency as “the ability of an AI system to be apparent and traceable in its functionality” (Liao et al., 2024).

Traceability is especially important in search because users rely on systems to provide accurate and verifiable information. However, the authors are cautious that “many current implementations of AI systems are opaque to end users” (Liao et al., 2024). This opacity creates uncertainty about how answers are generated, potentially undermining confidence in the technology. Traditional search engines by contrast, typically reveal sources directly through ranked links, allowing users to evaluate credibility themselves within those sources. Liao also establishes a strong connection between transparency and trust. He and his colleagues reason that explainable AI is “central to supporting trust, acceptance, and satisfaction with the end users that directly impact the system adoption” (Liao et al., 2024). This relationship suggests that the success of AI enhanced search will depend not only on performance but also on whether users feel they comprehend the system. Equally important is the warning that “lack of transparency and control decreases trust and user satisfaction” (Liao et al., 2024). This insight directly supports one of the central research questions of this thesis regarding whether AI systems sacrifice clarity for convenience. If users cannot see how results are produced, they may question their reliability even when the answers appear helpful. Beyond transparency, human AI interaction research highlights the importance of user experience as a design priority. Scholars note that “improving user experience makes AI systems more accessible to end users and aligns systems with human cognitive abilities” (Liao et al., 2024). Accessibility in this sense refers not only to usability but also to cognitive compatibility. Systems that click with how people think are more likely to be accepted. The authors further observe that guiding users and providing explanations “is an effective guideline for user experience” because it helps individuals build trust in automated decisions (Liao et al., 2024). This idea is particularly relevant when comparing AI generated answers with traditional search results. Explanatory features in these AI models may bridge the gap between algorithmic complexity and user understanding.

When viewed collectively, these studies suggest that the future of search technology depends heavily on user centered design principles. AI cannot simply produce answers. It must also communicate reasoning in ways that users can comprehend so those answers are validated. Circling back to consumer focused research, Kim and Priluck play up that technological adoption is shaped by psychological relationships. They argue that their work introduces “new psychological and relational dimensions to consumer search behavior” (Kim & Priluck, 2025). This contribution moves the discussion beyond functionality and into the realm of emotional and cognitive connection within users. Their research also highlights the importance of familiarity, explaining that repeated exposure leads individuals to develop “more favorable attitudes toward a stimulus with repeated exposure due to increased familiarity” (Kim & Priluck, 2025). This finding clearly explains why traditional search engines continue to dominate despite the emergence of AI alternatives. Users often trust what they know.

However, familiarity alone does not determine preference. The authors note that examining differences between chatbots and traditional engines provides “important and timely insights into the dynamics of consumer search behavior” (Kim & Priluck, 2025). These dynamics are precisely what this thesis seeks to understand through survey based research. The persistence of traditional search is echoed again in Hersh’s work. He reminds readers that people still search “to help answer direct questions or make decisions, or to use information for more integrative tasks, such as writing and teaching” (Hersh, 2024). This observation reinforces the idea that search is deeply immersed in intellectual activity. At the same time, Hersh points out a major limitation of generative AI outputs, explaining that varied information needs are “at odds with the output of generative AI chatbots that provide no or few references" (Hersh, 2024). Without references, verification becomes difficult, raising concerns about reliability and variability. He further stresses the ongoing challenge of information quality, noting that the emergence of ranking systems provided "a good if imperfect proxy for quality" (Hersh, 2024). This statement captures a key tension within search research. Even traditional methods are imperfect, yet they offer mechanisms for evaluation that AI responses sometimes disregard.

Motivation is widely recognized in the literature as a critical aspect. Liaw and Huang conclude that "motivation is a key factor that predicts individual intention to use search engines for information retrieval" (Liaw & Huang, 2011). This insight suggests that user behavior is shaped not only by technological capability but also by perceived usefulness and enjoyment. They also emphasize that a user with more experience tends to show "more positive attitude toward search engines as an information retrieval tool" (Liaw & Huang, 2011). This finding has important implications for AI adoption. If familiarity fosters positive attitudes, then widespread exposure to AI search tools may gradually shift user trust. Across these works, several intellectual inspirations surface clearly. Researchers in information retrieval support the recognition that search remains a foundational technology. Scholars in human computer interaction encourage a focus on usability and transparency. Consumer psychology research strongly encourages examination of trust and familiarity. Human centered AI frameworks push the conversation toward ethical and behavioral consequences. Together, this literature demonstrates that the evolution of search engines cannot be understood solely through technical metrics. Instead, it must be examined through the lens of human experience. Users are not passive recipients of information. They interpret interfaces, evaluate credibility, and form relationships with technologies.

This central idea directly supports the objectives outlined in the thesis introduction. Previous research has shown that traditional search engines provide structured access to information but may struggle with precision and response time. Emerging AI systems promise efficiency and conversational interaction but introduce concerns related to transparency, verification, and user understanding. Scholars have begun exploring these tensions, but significant gaps remain in comparative perception research. This thesis study builds upon these foundations by investigating how real users evaluate both models. In doing so, it responds to calls within the literature for more human centered analysis of AI technologies and contributes empirical insight into one of the most consequential shifts in modern information access. Ultimately, the reviewed literature reveals a shared understanding that search technologies shape how knowledge is constructed and trusted. The major influencers discussed throughout this review have collectively inspired an approach that treats search not merely as a computational process but also a human one. By extending this work through direct comparison of traditional and AI enhanced systems, this thesis aims to advance scholarly understanding of trust, usability, and information behavior in an increasingly AI mediated society.

III. METHODOLOGY AND PROCESS:

This study adopts a user-centered approach to examine how individuals perceive, trust, and interact with traditional search engines compared to AI-enhanced search systems. As established in the preceding literature review, prior research has demonstrated that search technologies are not only technical systems but also environments that shape how users interpret and evaluate information. While traditional information retrieval models emphasize structured access to sources and transparency, more recent AI-driven systems prioritize efficiency and conversational interaction, often at the expense of clear source visibility. These differences highlight the importance of examining not only how systems function, but how they are experienced by users. Building on these insights, the present study is designed to evaluate how variations in system design influence user perception, particularly in terms of trust, usability, and understanding of information.

To support a direct comparison between search approaches, this work is conducted alongside a complementary system-level implementation. A custom traditional search engine is developed using a shared Wikipedia dataset, providing a consistent comparison that reflects how traditional keyword-based search systems typically work, as described in earlier research. In addition to returning ranked documents, this system is also paired with an AI layer that processes those same retrieved results to generate summarized answers. Rather than relying on an external knowledge base, the AI operates directly on the output of the traditional system, producing responses that include synthesized explanations along with relevant documents and key points related to the query. This setup allows for comparison between two modes of interaction over the same underlying information, a traditional list-based retrieval approach and an AI-generated, answer-based interface. By ensuring that both traditional and AI-based responses are grounded in comparable information, the study creates a structured environment in which differences in user experience can be meaningfully evaluated. This design reflects the broader goal of the thesis, which is not to isolate technical performance alone, but to understand how those technical differences translate into user-facing outcomes.

The evaluation process is centered around a series of realistic search tasks that reflect common information-seeking behavior. These tasks are designed to mirror how individuals use search engines in everyday contexts, such as seeking explanations, comparing information, or answering specific questions. Participants are asked to interact with both the traditional system and AI-enhanced platforms while completing these tasks, allowing them to directly experience differences in how information is presented. Following this interaction, participants complete a structured survey designed to capture their perceptions of each system. This process ensures that responses are grounded in direct experience rather than abstract opinion, aligning with prior research that emphasizes the role of interaction in shaping user attitudes toward search technologies.

The survey serves as the primary instrument for data collection and is intentionally designed to measure dimensions of user experience that are consistently identified in the literature as critical to evaluating search systems. These dimensions include perceived accuracy, trustworthiness, usability, satisfaction with results, understanding of how information is generated, confidence in verifying information, and overall system preference. Each of these measures corresponds to themes established in prior research. For example, studies on human and AI interaction highlight the importance of transparency and explainability in building trust, while research on user behavior demonstrates that perceived accuracy often shapes system preference even when objective accuracy is uncertain. Similarly, usability and satisfaction are widely recognized as key drivers of adoption, and the ability to verify information is closely tied to concerns about information literacy and critical evaluation. By incorporating these dimensions into the survey, the study ensures that data collection is directly grounded in established research while remaining focused on the central objectives of this thesis.

In addition to quantitative measures, the survey includes open-ended questions that allow participants to explain their reasoning and describe their experiences in greater detail. This qualitative component is important because user perception is influenced not only by measurable factors but also by individual interpretation, familiarity, and expectations. Prior research suggests that attitudes toward search technologies are shaped by psychological and contextual factors, and the inclusion of open-ended responses provides insight into these influences. Together, the quantitative and qualitative elements of the survey create a more comprehensive understanding of how users experience different search systems.

The study incorporates a total of 150 survey responses, divided across three sources to balance control, realism, and variability. Fifty responses are generated through repeated trials conducted by the researchers, allowing for consistent evaluation under controlled conditions. Fifty additional responses are collected from AI undergraduate students who vary with their usage in search technologies for academic and everyday purposes. Due to practical limitations related to time constraints and accessibility, broader distribution of the survey to different UMass students was not feasible, and participant recruitment could not be conducted. While this approach limits the scope of generalization, it allows for focused and manageable data collection within the constraints of the study.

To further expand the range of perspectives represented, the fifty responses generated using an AI system are designed to simulate diverse user profiles. Each AI-generated response adopts the perspective of a distinct "student," varying in academic background, familiarity with technology, and experience with search systems. For instance, one simulated response may reflect the perspective of a first-year student with limited technical experience, while another may represent a senior computer science student who frequently engages with complex digital tools. This approach allows the study to approximate a broader spectrum of user experiences, reflecting insights from prior research that emphasize the role of familiarity and expertise in shaping user attitudes. At the same time, these simulated responses are interpreted with caution, as they are based on modeled behavior rather than direct human experience.

While this methodology provides a structured framework for comparison, it also includes certain drawbacks that should be considered when interpreting the results. The use of repeated evaluations and a relatively focused participant group may influence the consistency and scope of the findings, reflecting practical constraints in study design and data collection. In addition, while the inclusion of AI-generated responses helps introduce a broader range of perspectives, these responses are based on modeled behavior rather than direct human experience. These factors are common in exploratory, controlled studies of this nature and were necessary to maintain consistency across conditions. Future work could expand participant diversity and incorporate larger-scale data collection to further strengthen and generalize the findings. Despite these considerations, the overall approach remains effective for capturing meaningful patterns in how users perceive and interact with different search systems.

All things considered, this methodology is designed to directly address the central research objective by examining how differences between traditional and AI-enhanced search systems influence user perception in a controlled setting. By grounding the evaluation in both established research and direct interaction, the study connects technical differences in system design to meaningful user outcomes. This approach aligns with the more extensive goals of the thesis, which seeks to move beyond purely technical comparisons and instead provide a more integrated understanding of how evolving search technologies shape the way users interpret, trust, and engage with information.

IV. REPORT & RESULTS:

This section presents the findings from the user-centered evaluation of traditional and AI-enhanced search systems. While the technical performance of the custom search engine, including retrieval accuracy and ranking behavior, is explored in detail in Shahab's thesis, this section focuses on how those results are experienced and interpreted from a user perspective. The goal is not only to evaluate which system performs better, but to understand how users perceive, trust, and interact with each approach when completing realistic search tasks.

To support a structured and consistent comparison, a total of 150 responses were collected and analyzed. These responses were intentionally divided into three distinct groups: 50 responses from myself, 50 responses from Shahab, and 50 responses generated using a large language model. This separation was a key design decision in the evaluation process. Rather than combining all responses into a single dataset, dividing them allowed for a more detailed analysis of how evaluation differs depending on whether the evaluator is a human or an AI system.

This distinction is important because search systems are ultimately designed for human users. While AI-generated responses can simulate evaluation, they do not reflect real-world habits, familiarity, or prior experience. By separating human and AI responses, this study is able to capture both perspectives: how users actually behave when interacting with search systems, and how those systems might be evaluated from a more neutral, capability-focused viewpoint.

A. Overall System Preference:

One of the most consistent patterns observed in the data was a strong overall preference for traditional search systems. Across all responses, approximately 70% indicated a preference for traditional search, while a significantly smaller portion favored AI-enhanced systems or selected both equally. This result is notable because it contrasts with the increasing popularity and visibility of AI-powered search tools. While AI systems offer new capabilities such as direct answer generation and summarization, users still tend to favor systems that provide structured lists of results. This preference reflects more than just functionality. It highlights the importance of trust, familiarity, and control in shaping how users interact with information. As shown in

Figure 1, which compares responses between AI-based and traditional (human-preferred) search systems using a pie distribution, the difference in preference is clearly visible, with traditional search dominating the responses.

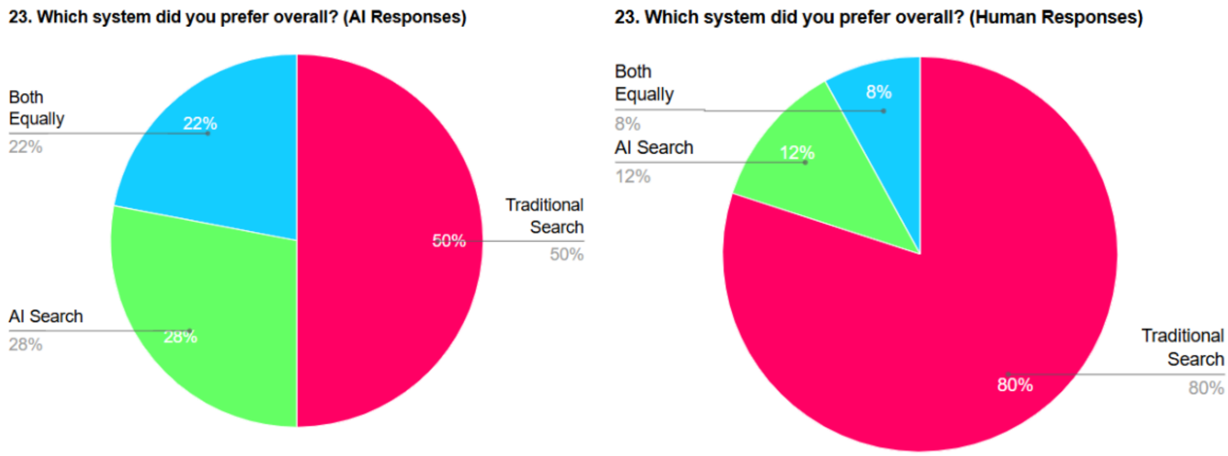

Figure 1: Human and AI Responses of overall preference of traditional vs AI enhanced search engines.

A closer analysis of the responses reveals several reasons for this preference. Many responses emphasized that traditional search allows users to see multiple sources at once, compare information across different websites, and make independent decisions about credibility. This process creates a sense of transparency, as users can clearly identify where information is coming from. In contrast, AI-generated answers often present information as a single, consolidated response. While this can be efficient, it also removes visibility into the underlying sources. Even when citations are included, users may still feel uncertain about how the answer was constructed. This uncertainty can reduce confidence in the results, even if the information itself is accurate. These findings align closely with the system-level observations described in Shahab's thesis. While the custom search system is capable of retrieving relevant documents and generating AI-based summaries, the way those results are presented plays a major role in how they are perceived. From a technical standpoint, the system may be functioning effectively. However, from a user perspective, transparency and control are just as important as accuracy.

B. Perceived Speed and Efficiency:

Another key area of evaluation focused on perceived speed and efficiency. Participants were asked which system helped them find information faster. The results showed that 52.9% of responses favored traditional search, while only 23.5% favored AI-enhanced search. The remaining responses indicated that both systems were equally effective. As shown in Figure 2, which presents user responses comparing perceived search speed between AI and traditional systems, traditional search was more frequently identified as faster overall.

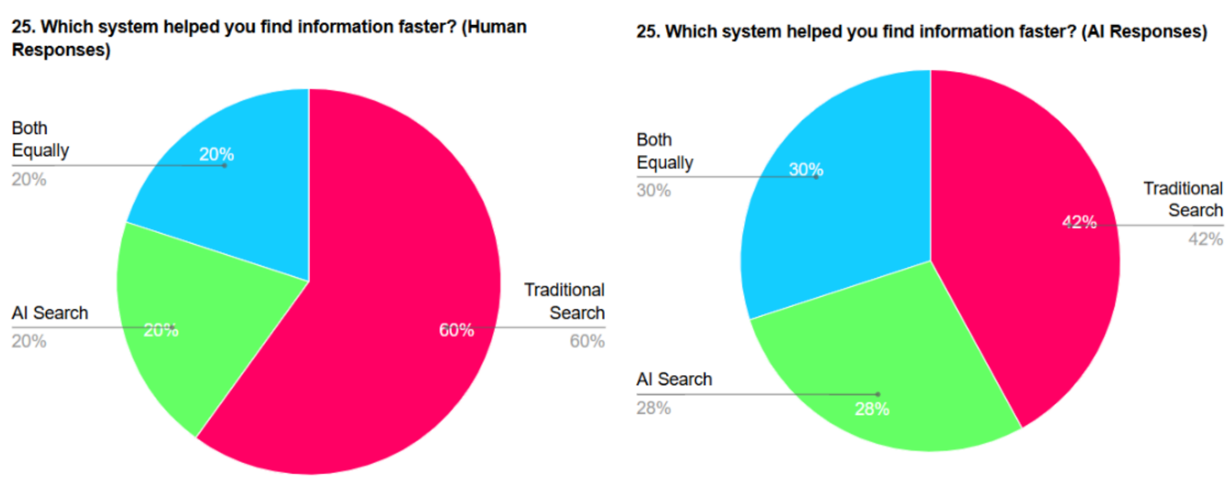

Figure 2: Human and AI Responses of speed of information retrieval of traditional vs AI enhanced search engines.

At first glance, this result may seem unexpected. AI systems are designed to provide quick, direct answers, reducing the need for users to click through multiple links. However, this finding highlights an important distinction between objective speed and perceived speed. Objective speed refers to how quickly a system generates a response. In this sense, AI systems often perform well, as they can produce a complete answer almost immediately. Perceived speed, on the other hand, depends on how quickly a user can interpret and trust that response.

Many responses suggested that traditional search feels faster because it follows a familiar structure. Users are accustomed to scanning titles, identifying keywords, and selecting relevant links. This process becomes efficient over time, as users develop strategies for navigating search results quickly. In contrast, AI-generated responses require users to process a block of text and evaluate its accuracy. This can introduce hesitation, particularly when the source of the information is not immediately clear. As a result, even though the system may provide an answer quickly, the overall experience may feel slower. This finding suggests that efficiency in search is not solely determined by system design. It is also influenced by user familiarity and confidence. A system that aligns with established user behavior may be perceived as more efficient, even if it involves additional steps..

C. Preference for Different Types of Queries:

While traditional search systems were preferred overall, the results show that this preference is not consistent across all types of queries. When participants were asked which system they preferred for quick, everyday questions, the distribution of responses shifted significantly. In these cases, approximately 41.2% of responses favored AI-enhanced search, compared to 29.4% for traditional search and 29.4% selecting both equally. This indicates that users recognize the advantages of AI systems in specific contexts, particularly when the goal is to obtain a quick answer rather than conduct in-depth research. As shown in Figure 3, which illustrates user preferences for quick, everyday queries across AI, traditional, and combined search systems, AI-based search becomes more competitive in this context compared to the overall results.

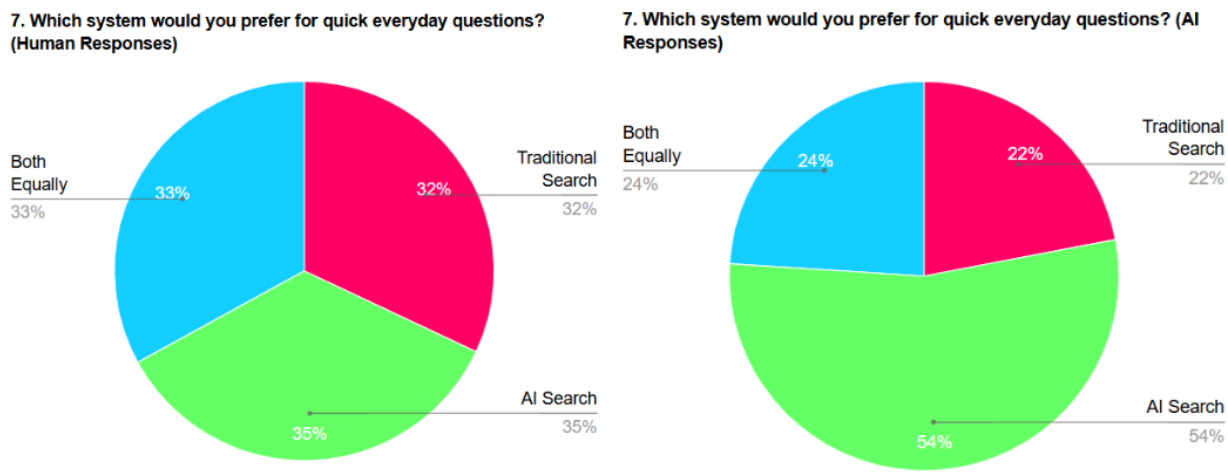

Figure 3: Human and AI Responses of preference for answering quick everyday questions of traditional vs AI enhanced search engines.

For simple factual queries, definitions, or general explanations, AI-generated responses provide a convenient and efficient solution. Users do not need to navigate multiple sources or synthesize information themselves. Instead, they can rely on the system to present a summarized answer. However, for more complex or detailed queries, the preference shifts back toward traditional search. In these scenarios, users value the ability to explore multiple perspectives, verify information, and engage more deeply with the content. This suggests that the effectiveness of a search system depends on how well it matches the user's intent. These findings reinforce the idea that search is not moving toward a single dominant model. Instead, users are adapting their behavior based on the task at hand. Traditional and AI-enhanced systems serve different purposes, and users choose between them depending on their needs.

D. Human vs AI Response Patterns:

A central component of this evaluation was the separation between human-generated and AI-generated responses. This distinction provides insight into how evaluation differs depending on the perspective of the evaluator.

By observing figures 1, 2, and 3 above it is apparent how human responses were more consistent and showed a stronger preference for traditional search systems. These responses often referenced familiarity, trust, and the ability to verify information as key factors in decision-making. Many responses reflected real-world habits, such as scanning search results and selecting sources based on prior experience. In contrast, AI-generated responses were more evenly distributed across answer choices. These responses were more likely to select “both equally” or to highlight strengths in both systems. This suggests that AI-generated evaluations are less influenced by experience and more focused on comparing system capabilities.

This difference highlights a fundamental distinction between experiential and theoretical evaluation. Human users evaluate systems based on how they actually use them, including their comfort level and prior knowledge. AI systems evaluate based on what the systems are designed to do, without the influence of habit or familiarity. This gap is important because it shows that system performance alone does not determine user preference. Even if an AI system offers advanced features, users may still prefer traditional search if it aligns more closely with their expectations and experience.

E. Alignment with System-Level Results:

When these user-centered findings are compared with the technical results presented in Shahab’s thesis, a clear pattern emerges. The system demonstrates reasonable retrieval performance based on standard metrics such as Precision@5 and nDCG@5. However, these metrics do not fully capture how users perceive the system. For example, a system may retrieve relevant documents according to benchmark metrics, but users may still feel uncertain if those results are presented in a way that lacks transparency. Similarly, AI-generated summaries may appear helpful, but users may question their reliability if they cannot easily verify the sources.

This highlights an important limitation of purely technical evaluation. Metrics such as precision and recall measure accuracy, but they do not account for trust, usability, or user satisfaction. By combining system-level analysis with user-centered evaluation, this study provides a more complete understanding of how search systems function in practice.

F. Summary of Key Findings:

Overall, the evaluation reveals several important insights. Traditional search systems remain the preferred option for most users, primarily due to familiarity, transparency, and the ability to verify information. Perceived speed is influenced by user experience, not just system performance. Traditional search is often viewed as faster because it aligns with established user behavior. AI-enhanced systems perform better in scenarios involving quick, simple queries, where convenience is prioritized over depth. Human and AI-generated responses differ significantly, reflecting experiential versus capability-based evaluation. User preference is context-dependent, with different systems being more effective for different types of tasks.

These findings suggest that the evolution of search is not a straightforward transition from traditional systems to AI-driven systems. Instead, it represents a shift toward a more flexible, hybrid model, where users select the approach that best fits their needs.

G. Discussion and Analysis:

The results of this study highlight a central idea that extends beyond simple system comparison: search technologies cannot be fully understood through technical performance alone. While Shahab's work provides important insight into how effectively the system retrieves and ranks documents, the user-centered findings presented in this thesis reveal that performance metrics do not directly translate into user preference, trust, or perceived usefulness. Instead, user interaction with search systems is shaped by a combination of familiarity, transparency, control, and confidence in the information being presented.

This section interprets the findings from both perspectives and explores what they reveal about the evolving role of search systems. By connecting system-level behavior with user perception, it becomes possible to better understand not only how search systems function, but how they are experienced in real-world contexts.

H. The Gap Between Performance and Perception:

One of the most important insights from this study is the clear gap between measured system performance and perceived usefulness. From a technical standpoint, the custom search system demonstrates reasonable effectiveness. As described in Shahab's thesis, the system is capable of retrieving relevant documents, ranking them using embedding-based similarity, and producing AI-generated summaries based on those documents. Standard information retrieval metrics such as Precision@5 and nDCG@5 provide structured evidence that the system is functioning as intended. However, the user-centered results show that this level of performance does not automatically lead to preference or trust. Even when the system retrieves relevant information, users may still favor traditional search systems. This suggests that performance alone is not the primary factor influencing user decisions.

Instead, users evaluate search systems based on how information is presented and how confident they feel in interpreting it. Traditional search engines provide a list of sources, allowing users to see where information comes from and to verify it independently. This creates a sense of transparency that supports trust, even if the results are not perfect. In contrast, AI-generated responses often present a single, consolidated answer. While this can improve efficiency, it also reduces visibility into the underlying sources. As a result, users may feel uncertain about the reliability of the information, even if it is technically accurate. This uncertainty highlights the importance of transparency as a key factor in user perception.

This gap between performance and perception demonstrates that improving search systems is not only a technical challenge, but also a design challenge. Systems must not only retrieve accurate information, but also present that information in a way that users can understand and trust.

I. Transparency vs Convenience:

A major theme that emerges from the analysis is the tradeoff between transparency and convenience. Traditional search systems prioritize transparency by providing multiple sources and allowing users to explore them independently. This approach requires more effort from the user, but it supports critical evaluation and informed decision-making. AI-enhanced systems, on the other hand, prioritize convenience by reducing the effort required to find information. By generating direct answers, these systems eliminate the need to navigate multiple sources. This can be especially useful for simple queries or situations where users need quick information.

However, this convenience comes at a cost. When users rely on AI-generated answers, they may not engage with the underlying sources or evaluate the credibility of the information. This can reduce their ability to critically assess what they are reading. The results of this study reflect this tradeoff clearly. Users preferred traditional search for tasks that require deeper understanding, verification, or confidence in the information. In these situations, transparency is more important than speed. At the same time, users were more willing to use AI systems for quick, everyday queries where convenience is the primary goal.

This suggests that the effectiveness of a search system depends on how well it balances these two factors. A system that provides both efficient answers and clear access to sources may be more effective than one that prioritizes only one aspect.

J. The Role of Familiarity and User Behavior:

Another key finding from this study is the strong influence of familiarity on user preference. Human responses consistently showed a preference for traditional search systems, even in cases where AI systems offered clear advantages in speed or simplicity. This suggests that user behavior is shaped not only by system capabilities, but also by long-term experience. Users have spent years interacting with traditional search engines, developing strategies for navigating search results, identifying relevant sources, and evaluating information. These habits create a sense of comfort and confidence that influences how new systems are perceived.

When users encounter AI-generated answers, they are interacting with a relatively new model of search. Even if the system provides accurate information, the lack of familiarity can create hesitation. Users may question how the answer was generated or whether important details have been omitted. This effect helps explain why traditional search remains dominant despite the rapid growth of AI-enhanced systems. It is not necessarily that traditional systems are always better, but that users are more comfortable with them. At the same time, the results suggest that this preference may evolve over time. As users become more familiar with AI systems, their perceptions may change. This highlights the importance of considering not only current behavior, but also how behavior may shift as technology becomes more widely adopted.

K. Human vs AI Evaluation: Experiential vs Theoretical Perspectives:

The separation between human-generated and AI-generated responses provides one of the most interesting insights in this study. The differences between these groups reveal two distinct ways of evaluating search systems. Human responses reflect an experiential perspective. These evaluations are based on real-world usage, including habits, preferences, and prior knowledge. Human participants consider factors such as trust, familiarity, and ease of use when evaluating systems.

AI-generated responses, on the other hand, reflect a more theoretical perspective. These evaluations focus on the capabilities of each system, such as speed, efficiency, and functionality. AI responses are more balanced and less influenced by prior experience. This difference is important because it highlights a limitation in how search systems are often evaluated. Technical and theoretical evaluations may suggest that a system is effective, but they do not fully capture how users will interact with that system in practice.

For example, an AI system may be capable of generating accurate and comprehensive answers, but users may still prefer traditional search if they feel more confident in verifying information themselves. This demonstrates that user experience cannot be fully predicted by system capabilities alone. Understanding this distinction is essential for designing effective search systems. Developers must consider not only what a system can do, but how it will be perceived and used by real users.

L. Task-Dependent Search Behavior:

One of the most crucial insights from the analysis is that user preference is highly dependent on the type of task being performed. The results show that users do not consistently prefer one system over the other. Instead, they choose different systems based on their specific needs. For simple, low-stakes queries, users are more likely to prefer AI-enhanced systems. These queries often involve straightforward questions that can be answered quickly without the need for detailed verification. In these cases, the convenience of AI-generated answers is highly valuable.

For more complex or high-stakes queries, users tend to prefer traditional search systems. These queries require deeper understanding, comparison of multiple sources, and greater confidence in the information. In these situations, the transparency of traditional search becomes more important. This behavior suggests that users are not replacing traditional search with AI, but rather incorporating AI into their existing workflows. Search is becoming more flexible, with users selecting the tool that best fits their needs. This finding supports the idea that the future of search is likely to involve a hybrid model. Instead of one system dominating, multiple approaches will coexist, each serving different purposes.

M. Implications for Search System Design:

The findings from this study have several important implications for the design of future search systems. First, they highlight the importance of balancing performance with user experience. Systems that focus solely on improving accuracy or speed may not be effective if they do not also address user trust and understanding. Second, the results suggest that transparency should remain a key design priority. Even as AI systems become more advanced, users still value the ability to see and verify sources. Providing clear connections between generated answers and underlying documents may help bridge the gap between convenience and trust. Third, the study highlights the importance of designing systems that adapt to different types of queries. Rather than using a single approach for all tasks, search systems could dynamically adjust their behavior based on user intent. For example, a system could provide a direct answer for simple queries while also offering detailed source lists for more complex questions. Finally, the differences between human and AI evaluation suggest that user testing should play a central role in system development. Technical performance metrics are important, but they must be complemented by user-centered evaluation to ensure that systems meet real-world needs.

N. Limitations of the Study:

While this study provides meaningful insights, it is important to acknowledge its limitations. The evaluation was conducted as a controlled proof-of-learning project, with responses generated by a limited number of participants and an AI system. This limits the extent to which the results can be generalized to a broader population. Furthermore, the inclusion of AI-generated responses introduces a unique perspective, but it does not fully replicate real user behavior. While these responses are useful for comparison, they should be interpreted as a complement to human evaluation rather than a replacement.

The study also focuses on a specific dataset and system implementation. While this allows for a controlled comparison, it does not capture the full complexity of real-world search systems. Despite these limitations, the study achieves its primary goal of exploring how different search systems are experienced and evaluated. By combining system-level analysis with user-centered evaluation, it provides a more complete understanding of the tradeoffs involved in modern search.

O. Overall Interpretation:

Taken together, the findings from this study suggest that the evolution of search is not a simple transition from traditional systems to AI-driven systems. Instead, it represents a shift toward a more flexible and user-driven model of information retrieval. Traditional search systems continue to play an important role due to their transparency, familiarity, and support for critical evaluation. AI-enhanced systems introduce new capabilities that improve efficiency and accessibility, but they also change how users interact with information.

The most important takeaway is that search is not just about retrieving information. It is about how that information is presented, interpreted, and trusted. Understanding this relationship is essential for developing systems that not only perform well, but also support meaningful and confident user interaction.

V. CONCLUSION:

This study set out to explore how traditional and AI-enhanced search systems differ not only in terms of technical performance, but more importantly in how they are perceived, trusted, and used by individuals in real-world contexts. While Shahab's work focused on the design, implementation, and system-level evaluation of a custom search engine, this thesis complements that perspective by examining how users interact with and interpret the outputs of these systems. By combining these two approaches, this project provides a more complete understanding of modern search, bridging the gap between how systems function and how they are experienced.

At a system level, the custom search engine developed alongside this study demonstrates that it is possible to construct a controlled retrieval environment that produces meaningful results using a shared dataset. Through embedding-based retrieval, structured ranking, and the integration of an AI-generated overview, the system reflects many of the key characteristics of modern search engines. It highlights the shift from traditional document retrieval toward hybrid systems that combine retrieval with generation. This technical foundation is essential because it ensures that both traditional and AI-enhanced outputs are grounded in the same underlying data, allowing for a fair and consistent comparison.

However, the primary contribution of this thesis lies in examining how these outputs are perceived by users. The findings reveal that traditional search systems remain the preferred option in most cases, particularly when users are seeking reliable, verifiable, and detailed information. This preference is not solely based on system performance, but on how information is presented and how users engage with it. Traditional search allows users to view multiple sources, compare different perspectives, and make independent decisions about credibility. This interaction model supports a sense of control and transparency that users continue to value. In contrast, AI-enhanced search systems introduce a fundamentally different approach to information access. By generating direct answers, these systems reduce the effort required to find information and streamline the search process. This can be highly effective for simple or everyday queries, where users prioritize speed and convenience over depth and verification. The results of this study reflect this distinction clearly. While traditional search dominates overall preference, AI systems perform more strongly in scenarios where quick answers are sufficient.

This pattern suggests that the evolution of search is not a simple transition from one system to another, but rather the emergence of a more flexible and task-dependent model. Users are not abandoning traditional search in favor of AI, nor are they rejecting AI entirely. Instead, they are adapting their behavior based on the context of their information needs. Traditional search remains essential for tasks that require exploration and verification, while AI-enhanced search is used for efficiency and convenience. This hybrid approach represents a significant shift in how search is understood and utilized. A central insight from this study is the distinction between system performance and user perception. While Shahab's system-level analysis demonstrates that the search engine performs reasonably well according to standard information retrieval metrics, these metrics do not fully capture how users evaluate the system. Metrics such as precision and recall provide valuable information about accuracy, but they do not account for factors such as trust, usability, and confidence in the results. This gap highlights the importance of incorporating user-centered evaluation into the study of search systems.

Users do not interact with search systems as abstract algorithms. They experience them as tools that either support or hinder their ability to find and understand information. As a result, their evaluations are influenced by factors that extend beyond technical performance. Familiarity, transparency, and control all play a significant role in shaping how users perceive different systems. Even when AI-generated answers are accurate, users may hesitate to trust them if they cannot easily verify the information or understand how it was produced. This hesitation is closely tied to the concept of transparency. Traditional search systems provide clear visibility into the sources of information, allowing users to trace results back to their origins. AI-enhanced systems, on the other hand, often present information in a synthesized form, making it more difficult to identify the underlying sources. While this approach improves efficiency, it can also reduce confidence, particularly in situations where accuracy is critical.

The separation between human-generated and AI-generated responses further reinforces this idea. Human responses were strongly influenced by real-world experience and familiarity with traditional search systems. These responses consistently emphasized the importance of being able to verify information and access multiple sources. In contrast, AI-generated responses were more evenly distributed and often highlighted the strengths of both systems. This difference reflects two distinct perspectives on evaluation. Human evaluations are experiential. They are shaped by how users actually interact with systems, including their habits, expectations, and comfort levels. AI evaluations in contrast, are more theoretical focusing on system capabilities rather than user experience. This distinction is important because it highlights a gap between how systems are designed and how they are used. A system may be technically advanced, but if it does not align with user expectations, it may not be preferred.

Another important takeaway from this study is the role of familiarity in shaping user behavior. Traditional search engines have been a central part of the digital experience for decades, and users have developed well-established strategies for interacting with them. These strategies allow users to navigate search results efficiently and with confidence. AI-enhanced systems, while powerful, represent a newer interaction model. As a result, users may be less comfortable relying on them, even when they provide useful information. This suggests that user preference is not static, but may evolve over time as familiarity with AI systems increases. As users become more accustomed to AI-generated answers, their trust in these systems may grow. However, this transition will likely depend on how well these systems address concerns related to transparency and reliability. Simply improving performance may not be enough. Systems must also be designed in a way that supports user understanding and confidence.

A. Future Directions:

While this study provides valuable insight into the relationship between traditional and AI-enhanced search systems, it also highlights several areas for future work that could build on these findings. One important direction is the expansion of the user study to include a larger and more diverse group of participants. Due to practical constraints, including time limitations and the absence of IRB approval, this study relied on responses generated by the researchers and an AI system. While this approach allowed for a controlled and consistent evaluation, future research could benefit from incorporating a broader range of participants. This would provide a more comprehensive understanding of how different users interact with search systems and whether the patterns observed in this study hold across real and different populations.

Another potential area for future work involves improving the integration between retrieval and generation within the search system. While the current implementation successfully uses retrieved documents to generate AI summaries, there is still room for improvement in how these components interact. Future systems could explore more advanced methods for ensuring that generated answers remain closely tied to their source documents. This could include stronger grounding techniques, improved citation methods, or interface designs that make it easier for users to trace information back to its origin.

In addition, future research could examine how user behavior changes over time as AI-enhanced search systems become more widely adopted. This study captures a snapshot of current user preferences, but these preferences may evolve as users gain more experience with AI tools. Longitudinal studies could provide valuable insight into how trust, efficiency, and interaction patterns develop with continued exposure to AI-driven search. Another important direction involves exploring the impact of search systems on information literacy and critical thinking. As AI-generated answers reduce the need for users to engage with multiple sources, it becomes important to understand how this affects their ability to evaluate information. Future studies could investigate whether reliance on AI summaries influences how users assess credibility, compare perspectives, and form conclusions.

Finally, future work could expand the technical capabilities of the system itself. While the current system provides a controlled environment for comparison, incorporating larger datasets, hybrid retrieval methods, or real-time data sources could make the system more representative of real-world search engines. This would allow for a more comprehensive evaluation of how different approaches perform in more complex and dynamic environments.

B. Closing Perspective:

At its core, this project represents a proof of learning through direct application. By building, testing, and evaluating both traditional and AI-enhanced search systems, this study moves beyond theoretical discussion and into practical exploration. It demonstrates that understanding search requires more than just knowledge of algorithms and metrics. It requires an understanding of how people interact with information and how technology shapes that interaction. The discoveries of this study suggest that the future of search will not be defined by a single system or approach. Instead, it will be shaped by the relationship between users and the tools they use to access information. As AI continues to transform search, the challenge will not only be to improve performance, but to ensure that these systems support trust, transparency, and meaningful user engagement.

In this sense, the evolution of search is not just a technical progression. It represents a deeper shift in how we come to understand the world around us. The way information is accessed, interpreted, and ultimately trusted shapes not only what we know, but how we think, how we question, and how we make decisions in our everyday lives. When answers are handed to us, the process of seeking, exploring, and discovering begins to change, and with it, the role we play in forming our own understanding. The way information is presented is no longer just a design choice. It becomes a defining factor in how knowledge is experienced and internalized. Recognizing this connection means recognizing that search systems are not just tools for finding information, but influences how individuals learn, think critically, and engage with the world. As these technologies continue to evolve, the goal should not only be to deliver faster or more efficient answers, but to support a deeper sense of awareness, curiosity, and confidence in the way we come to know what we know.

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